Midterm 2018, "The Mathematical Foundations of Political Methodology"

You have 120 minutes for this exam. You may use a calculator and refer to your Pemberton & Rau book. Please do not use the internet during the exam. Show your work.

1. Consider the following sets of real numbers:

$$\begin{aligned} A &= \{x : 1 \leq x \leq 5\} \\ B &= \{x : 0 \leq x \leq 2\} \\ C &= \{x : x < 0\} \end{aligned}$$

   Describe each of the following sets of real numbers in words

   (a) A^c

   The numbers greater than 5 and less than 1.

   (b) $A \cup B$

   The numbers between 0 and 5 inclusive.

   (c) $B \cap C^c$

   B, or the numbers between 0 and 2 inclusive.

   (d) $(A \cup B) \cap C$

   none.

   (e) $(A \cup B)^c \cap C$

   C, or the numbers less than 0.

2. Find the limit of each of the following sequences as $n \to \infty$, if the limit exists.

   (a)
$$\frac{4n}{(n^2 - 7)}$$

   The limit is 0

   (b)
$$\frac{4n^{\frac{1}{2}}}{(n^2 - 7)} + 1$$

   It's 1.

   (c)
$$\frac{4n}{(n^{\frac{1}{2}} - 7)} + 1$$

It doesn't exist.

3. Use the (δ, ϵ) definition of a limit to prove that

$$\lim_{x \to 3} 4x - 11 = 1$$

Proof:

For any $\epsilon > 0$, we can find a $\delta > 0$ such that $|4x - 11 - 1| < \epsilon$ whenever $|x - 3| < \delta$.

Rearranging, we get $4|x - 3| < \epsilon$. $|x - 3| < \frac{\epsilon}{4}$

Setting $\delta = \frac{\epsilon}{4}$ proves the result. $\square$

4. Use the definitions of continuity and differentiability to evaluate whether or not the following function is 1) continuous and 2) differentiable at $x = 0$.

This function is differentiable everywhere except at x=0, so it is continuous at all those points. The only question is whether or not it is continuous at x=0.

$$f(x) = x^{\frac{1}{3}}$$

The function is continuous at $x = 0$ if $lim_{x \to 0} f(x) = f(0)$. We can show it analytically using the reasoning from above, For any $\epsilon > 0$, we can find a $\delta > 0$ such that $|x^{\frac{1}{3}} - 0| < \epsilon$ whenever $|x - 0| < \delta$.

If we set $\delta = \epsilon^3$, we are set. $\square$

However the function is not differentiable at 0 because it would require us to divide by zero:

$$f(x) = \frac{1}{3} x^{\frac{-2}{3}}$$

5. Let

$$f(x) = \frac{x}{1 + e^x}$$

(a) Is f(x) concave, convex or both?

$$f'(x) = \frac{1 - e^x (x - 1)}{1 + e^x}^2$$

$$f''(x) = \frac{2xe^{2x}}{(1 + e^x)^3} - \frac{xe^x}{(1 + e^x)^2} - \frac{2e^x}{(1 + e^x)^2}$$

Plug in any small number, like 0.

$$f''(0) = -\frac{2e^0}{1+e^{0^2}} = -\frac{2}{2} = -1 < 0$$

However, plug in 5, you get a positive number

$$\frac{10e^{10}}{(1+e^5)^3} - \frac{5e^5}{(1+e^5)^2} - \frac{2e^5}{(1+e^5)^2} = .02$$

This function is positive in some bits and negative in others so it is both.

(b) Find any asymptotes that f may have.

The function approaches y=x on the left and an asymptote at the right at the x-axis.

6. Perform the following matrix operations:

(a) Find the inverse of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 & 1 & 0 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 1 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 1 \\ 0 & 0 & 5 & -3 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 1 \\ 0 & 0 & 1 & -\frac{3}{5} & \frac{2}{5} & \frac{3}{5} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & \frac{3}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & \frac{1}{5} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & -\frac{3}{5} & \frac{2}{5} & \frac{3}{5} \end{bmatrix}$$

(b) Find the rank of

$$A = \begin{bmatrix} 3 & 1 & 1 & 2 \\ 2 & 1 & -1 & 2 \\ 0 & 3 & 2 & 0 \\ 1 & 0 & 3 & 0 \end{bmatrix}$$

It is 4

(c) What x would make the following matrix singular?

$$\begin{bmatrix} 1 & 3 \\ x & 2 \end{bmatrix}$$

$1 * 2 - 3x = 0, \rightarrow \; x = 2/3$

7. If A and B are matricies, which of the following imply the others:

   (a) AB=BA

   (b) Either A or B is the identity matrix.

   (c) A is the inverse of B.

   (d) A and B are symmetric.

   c implies a, b implies a, a and d imply nothing else.

8. What constant k solves the following equation?

$$\int_0^{10} ky^2 dy = 1$$

$$k/3y^3|_{10} - k/3y^3|_0 = 1$$

$$k/3 * 10^3 = 1$$

$$k = 3/1000$$

   $k = 3/1000$

9. What values of the constant a satisfies the following equation?

$$\int_0^a \frac{2x}{1+x^2} dx = 1$$

$$U = 1 + x^2$$

$$du = 2x$$

$$log(1 + x^2)|_0^a$$

$$log(1 + a^2) - log(1) = 1$$

$$1 + a^2 = exp(1)$$

$$a = \sqrt{(exp(1) - 1)}$$

$$a = -\sqrt{(exp(1) - 1)}$$